Software Issues

Q: Why can't my compiler find the corresponding device?

- A: You need to build a development environment and install the corresponding project library before you can develop a device.

1 About myStudio

Q: What is myStudio?

- A: It is a self-developed software of our company. It is a tool for burning or modifying the firmware of the existing robotic arms launched by our company. More

Q: Is there any difference between the firmware used by MyCobot 320 PI-2020 and 2022 models?

A:

- The firmware requirements for earlier versions of myCobot 320 PI are as follows:
  - Ubuntu 18.04
  - atommain4.1
- The firmware requirements for the new version of myCobot 320 PI are as follows:
  - Ubuntu 18.04
  - picomain1.0
  - atommain5.0

The highest version of this firmware combination is mystudio v3.5.7

Q: Why can't the device run normally after the firmware is burned into the ATOM terminal?

- A: The firmware of the ATOM terminal must use our factory firmware. Other unofficial firmware cannot be changed during use. If the device is accidentally burned with other firmware, you can use myStudio to re-burn the ATOM firmware.

Q: Can the drag teaching in the firmware record the gripper movements?

- A: It is not possible to use drag teaching to record the gripper movements at the moment, because the gripper belongs to joint number 7, and our drag teaching can only record and play the movements of joints numbered 1-6.

Q: Why can't I drag and teach after burning the minirobot firmware?

- A: First check whether the M5Stack-basic firmware and atom firmware are burned, whether the burned firmware corresponds to the requirements to be implemented and whether the burned firmware is the latest version. It is recommended to burn the minirobot firmware to version v2.1 and the top atommain firmware to version v4.1 and above (need to support mystudio version v4.3.1 and above).

Q: What should I do if myStudio cannot recognize the serial port of myCobot?

- A: If your computer device does not prompt that it is connected to myCobot, please install the serial port driver first. It should also be noted that the Raspberry Pi, Arduino and Jetson nano series robotic arms **cannot be connected to a laptop using a data cable** , and the firmware needs to be burned using mystudio in the built-in system.

Q: Can the trajectory recorded by dragging and teaching be saved in the card?

- A: Currently, it is not possible to save to the memory card. And drag teaching can only save one path at a time, and the next recording will overwrite the previous action.

2 About RoboFlow

Q: Can I use robotStudio software programming?

- A: Our own industrial programming software roboFlow can be used. RobotStudio is from ABB and cannot communicate with us.

Q: What is the reason why the roboFlow software moves too quickly and exceeds the limit?

- A: It may be that one or more joints exceed the limit.

Q: How does RoboFlow load already written programs?

- A: After logging in, select Program Robot and then click Load Program. You cannot directly click to run the program, you can only click pro600.

3 About myCobot phone controller

Q:What version of firmware should be burned into the myCobot phone controller app?

- A: You need to burn the atom firmware atommain2.5 version in mystudio.

  > As of 2/4/2023, the mobile APP control function has been disabled. Please pay attention to the information release of the mystudio firmware version for the restart time.

4 About myblockly

Q: What is myblockly?

- A: [About myblockly]

Q: Why do pop-up boxes always appear when running myblockly?

- A: Before running the myblockly program, please close the serial port occupation.

Q: When myblockly is running, child process exited with code 0 appears. Why? ?

- A: This is not an error. The real error needs to be analyzed one by one. This string of characters represents the end of the program and returns the binary number 0, indicating termination.

5 About ROS1

Q: When I switch the terminal to `~/catkin_ws/src` and use git to install and update `mycobot_ros` , the target path "mycobot_ros" seems to already exist. Why is this?

- A: There is already a `mycobot_ros` package in `~/catkin_ws/src` , so you need to delete it first and then re-run git.

Q: When running rosrun, the terminal reports an error message `counld not open port /dev/ttyUSB0: Permission: '/dev/ttyUSB0'` . Why does this happen?

- A: Insufficient serial port permissions. Enter `sudo chmod 777 /dev/ttyUSB0` to grant permissions.

Q: Why can't the ROS program run in vscode?

- A: Since the vscode terminal cannot be loaded into the ros environment, it needs to be run on the system terminal.

Q: When running rosrun, the terminal prompts Unable to register with master node [http://localhost:11311]: master may not be running yet. Will keep trying. What is the reason?` ?

- A: Before running the ros program, you need to open the ros node and enter roscore in the terminal.

Q: When rosrun is running, the terminal displays an error message `counld not open port /dev/ttyUSB0: No such file or directory: '/dev/ttyUSB1'` . Why?

- A: The serial port is incorrect. You need to confirm the actual serial port of the current robot. You can check it through `ls /dev/tty*` .

Q: In Ubuntu 18.04, catkin_make fails to build the code, and the terminal prompts `Project 'cv_bridge' specifies '/usr/include/opencv' as an include dir, which is not found.` and other error messages

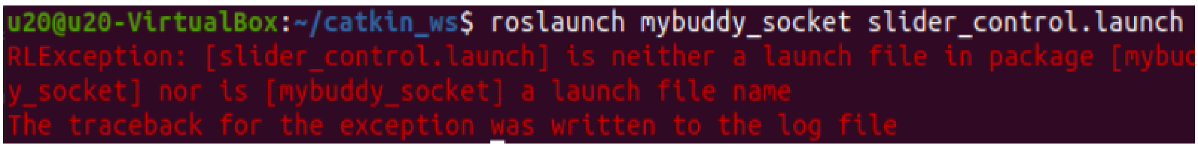
- A: The opencv path in the configuration file does not match the actual system path. You need to use sudo to modify the configuration file (the path is `/opt/ros/melodic/share/cv_bridge/cmake/cv_bridgeConfig.cmake`). The actual opencv path of the system is under `/usr/include/` .

Q: I just cloned the mycobot_ros package and then ran the rosrun program directly. But an error like `package 'mycobot_280' not found` or "the file cannot be found" occurred?

A: The newly cloned mycobot_ros needs to build code for ros environment compilation. Enter in the terminal

```
cd ~/catkin_ws/
catkin_make
source devel/setup.bash
```

Q: After compiling, why does the following error appear when running the launch command in a new terminal?



- A1: The system does not add ros environment variables, so you need to source it every time you open a new terminal:

```
cd ~/catkin_ws/
source devel/setup.bash
```

- A2: The system adds ros environment variables, and you don’t need to execute source every time you open a new terminal:

```
# The noetic is Ubuntu20.04 system
echo "source /opt/ros/noetic/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

- A3: The file name in the instruction may be inconsistent with the actual file name in the mycobot_ros package. Please check the instruction carefully for errors.